Detection of Fake Facial Images Using Advanced AI Techniques

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***Abstract*—The advent of advanced AI models capable of generating and manipulating facial images has led to significant ethical and security concerns. Fake face images, created through sophisticated techniques like Generative Adversarial Networks (GANs), pose challenges in digital identity verification, social trust, and biometric security systems. To address these challenges, this study benchmarks Vision Transformer (ViT), Compact Convolutional Transformer (CCT), and DinoV2 with custom layers for self-supervised learning and classification. Using a highly balanced dataset comprising 230k facial images exclusively of faces, our models aim to robustly distinguish between real and manipulated images. Additionally, we trained the DCGAN model on real face images from our training and utilized its discriminator to classify real and fake images. The proposed methodology enhances detection accuracy and paves the way for improved safeguards against the misuse of AI technologies, contributing to harassment prevention and security in face biometrics.**

I. Introduction

Artificial Intelligence (AI) has revolutionized the generation and manipulation of digital images, particularly facial images. Advanced AI models, including Generative Adversarial Networks (GANs), can now generate hyper-realistic fake faces [1], manipulate existing images, or seamlessly replace faces within an image. While these technologies showcase the immense potential of AI in the creative and entertainment industries, they also produce alarming misuse scenarios. For example, manipulated images can be weaponized to harass individuals, particularly by replacing faces with inappropriate or harmful content. Applications of detecting fake or manipulated images extend to social media integrity, law enforcement, digital identity verification, and biometric security systems, where it is crucial to determine the authenticity of facial data.

Detecting manipulated or fake facial images is increasingly challenging due to the sophistication of modern image generation techniques. Traditional detection methods, such as Local Binary Patterns (LBP) [2] and Error Level Analysis (ELA) [3], are inadequate against the realism of images created by advanced GAN architectures. Moreover, convolutional neural networks (CNNs), the standard for image classification, struggle to generalize well against diverse manipulation methods [4]. The rapid advancement of GAN models necessitates more robust detection mechanisms that can adapt to various forms of image alterations, making the problem a moving target for researchers.

Notably, the Deep Convolutional GAN (DCGAN) architecture [5] has emerged as a prominent unsupervised model capable of learning hierarchical representations of image data. This architecture demonstrates how unsupervised learning can generate realistic images and highlights the challenges in distinguishing real from fake images. Incorporating insights from DCGANs strengthens our understanding of GAN-generated manipulations and the need for advanced detection techniques.

Despite the growing prevalence of AI-generated fake images, most existing research relies on traditional or moderately advanced classification models that fail to achieve high accuracy and generalizability. A notable gap exists in leveraging self-supervised learning, which has the potential to uncover nuanced patterns in data without requiring extensive labeled datasets. Furthermore, the integration of custom layers tailored for manipulation detection remains underexplored. By addressing these gaps, more effective solutions can be developed to counteract the misuse of AI-generated content.

This paper compares state-of-the-art transformer-based architectures for detecting real and fake facial images. Specifically, Vision Transformer (ViT) [6] and Compact Convolutional Transformer (CCT) [7] are employed for classification tasks, while DinoV2 [8] is integrated with custom layers to leverage self-supervised learning and enhance detection performance. Additionally, the DCGAN [5] model is trained on real face images, and its discriminator is employed to classify real and fake images. Our contributions include:

- Comparing the performance of multiple advanced architectures for manipulation detection.
- Demonstrating the effectiveness of self-supervised learning in enhancing real-fake image detection.
- Addressing ethical and security concerns by providing insights to mitigate harassment risks and fortify facial biometric systems.

II. Literature Review

To enhance DeepFake images by eliminating detectable traces and making them appear more authentic, authors present a novel anti-forensics attack called the trace removal attack (TR-net) [9]. This attack can bypass multiple detection systems (spatial detection, frequency detection, fingerprint detection) without prior knowledge of their specific designs. The authors conducted experiments in heterogeneous security scenarios while maintaining high visual quality. To address various representational traces, the authors use adversarial learning with one generator and multiple discriminators. Here, the

authors point out the weaknesses of present-day detectors and highlight the requirement for more robust and generalized techniques for the future.

To provide an immense overview of digital image forgery detection techniques [10], the authors focus on altering images to conceal information and the importance of authenticity. In this paper, the author reviews various techniques like pixel-based, physics-based, geometry-based, camera-based, and format-based methods. The motives of those methods are splicing, copy-move, retouching, multiple JPEG, JPEG blocking, lighting environment, color filter array, and geometry. To improve the detection technique, the authors accentuate the significance of pre-processing approaches like Histogram Equalizations, Contrast Enhancements, and Median Filters. They emphasize the challenges of detecting complex forgery types, including hybrid approaches and compressed images.

Marina Gardella proposed the forensics similarity graphs method for image forgery detection [11]. In their methods, vertices represent image patches and edge weights reflect forensic similarity. The authors use two clustering methods (spectral clustering and modularity optimization) for detecting and localizing image forgeries and provide an online executable version for users to test on their images. The authors shift to a scalable and generalizable framework for forgery detection by exploiting local inconsistencies in image traces.

Ping Chen and Ming Xu suggest a Two-Branch Convolutional Network (TCNSC) model that learns two features [12]: compressed classification and similarity learning, to overcome the difficulties in identifying compressed DeepFake images. This method uses the intrinsic similarity between compressed and raw images—which is unaffected by compression factors—to extract reliable features. Based on experimental results, TCNSC achieves accuracies of 91.8%, 93.4%, and 95.3% for low, medium, and high-quality photos, respectively, showing a significant improvement over state-of-the-art techniques across a range of compression settings. The significance of compression-resilient detection and the demand for more reliable approaches to managing real-world situations are accentuated by the writers.

The authors introduce a Dual Input Convolutional Neural Network (DICNN) to escalate the detection of fake face images [13], which merges various input layers to bring out deep texture patterns. To provide transparency in decision-making, the models also use explainable AI techniques (SHAP) to achieve high accuracy. Despite other methods, this method performs better and also focuses on manipulated image detection. To gain the belief of security and forensic experts, this method emphasizes the integration of interpretability into DeepFake forensics.

In this paper [14], the authors investigate image authenticity detection using Error-Level Analysis (ELA) and Noise Analysis techniques. The authors discuss three manipulation techniques: splicing imagery, copy-moving images, and retouching imagery. The researchers demonstrate practical applications of forensic tools like ForensicallyBeta for analyzing JPEG images to determine the purity and authenticity of imagery. Together with other methods, such as Metadata and RGB Histograms, they can see the difference between the original image and the manipulated image to check the authenticity of an image. The result is that using ForensicallyBeta or other methods determines the authentic imagery.

To tackle the issues related to detecting digital image forgery, Nagarathna C R examines [15] various techniques that concentrate on copy-move, splicing, and removal forgery detection. The paper discusses pre-processing methods such as compression, normalization, and resizing, along with feature extraction techniques including SURF, CNNs, and SIFT. Classification techniques like SVMs and ensemble methods are also explored. The authors point out shortcomings like the lack of transparency in detection models, scarcity of datasets, and difficulties in assessing various types of forgeries. They stress the importance of developing robust machine-learning methods for producing generalizable and precise forgery detection.

Examining developments in multimedia forensics, [16] this overview focuses on spotting and detecting picture and video forgeries, including generative media. The authors address counter-anti-forensics tactics while pointing out how vulnerable current detection systems are to anti-forensic approaches. The study divides forgery detection into three categories: generating media detection, manipulation detection, and source identification. The various datasets and methodologies are compared in depth. The study highlights important gaps and calls for improvements in automated and real-time forensic systems, notwithstanding the lack of experimental validation. The difficulties presented by generative adversarial networks (GANs) are highlighted in this article, along with the necessity of new, scalable, and flexible detection methods.

[17]This study looks at cutting-edge deep learning methods like CNNs, Xception networks, and LSTMs for identifying deepfake media. It draws attention to the ethical and security issues raised by the quick growth of AI in manipulating multimedia for malevolent ends. The suggested investigation highlights the part adversarial networks play in the production of deepfakes by contrasting several techniques for identifying faces produced by GANs. Despite being theoretical, the study provides a basis for creating better detection systems. In order to improve model accuracy, particularly in harsh scenarios like compression and scaling, the work also emphasizes the necessity of domain knowledge and datasets.

In order to detect counterfeit digital photos, [18] in this study the authors present an enhanced DenseNet121-based architecture. In order to guarantee high accuracy and effective learning, so they provide a densely connected framework that tackles problems like gradient vanishing and unnecessary layer dependencies. With a 92.32% detection accuracy, the model beats traditional CNNs and other deep learning frameworks when tested on the CASIA V2.0 dataset. The study recommends DenseNet's adaptation for upcoming forensics difficulties while highlighting the benefits of its compactness and effective parameter utilization. Its restricted testing on a single dataset, however, allows for the assessment of its generalisability to a wider range of datasets and media formats.

III. Dataset and Model

A. Dataset

Our dataset is collected from Kaggle, authored by Shivam Ardeshna, and available at Kaggle [19]. The dataset contains four different sub-datasets, each comprising two classes: Real and Fake. In total, the dataset includes 230k facial images, with an equal distribution between the two classes. The dataset is divided into three sections: train, test, and validation, ensuring balanced representation across all subsets. The real class contains authentic, unaltered images, while the fake class consists of images manipulated or generated to represent various forms of digital forgeries.

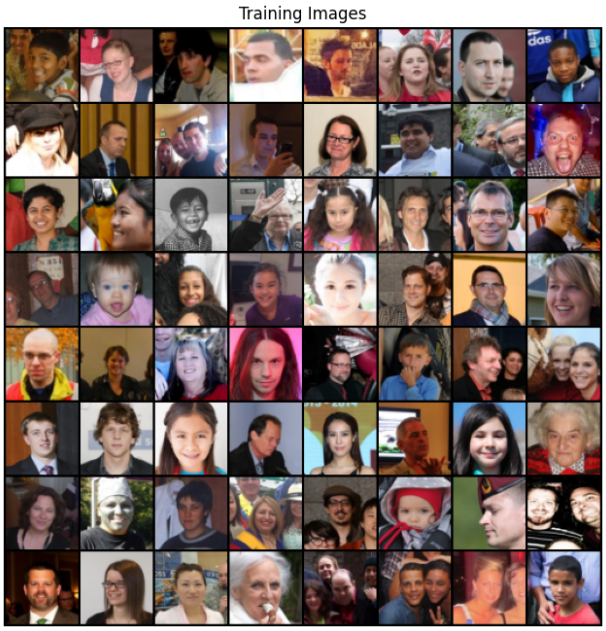


Fig. 1. Dataset Samples

B. Models

1) Vision Transformer (ViT): Vision Transformer (ViT) is a state-of-the-art architecture pre-trained on the ImageNet dataset. We customized the final classification layer to adapt to our dataset, allowing it to distinguish between real and fake images effectively. ViT's transformer-based approach processes image patches and computes relationships between them, providing robust feature extraction capabilities.

2) Compact Convolutional Transformer (CCT): Compact Convolutional Transformer (CCT) is another transformer-based model pre-trained on ImageNet. Like ViT, we customized its final layer to classify images from our dataset. CCT leverages convolutional layers to improve the processing of spatial information, making it particularly suitable for image classification tasks.

3) DinoV2: DinoV2 is utilized as a feature extractor, with pre-trained weights offering robust visual representations. Custom layers were added to enhance its classification performance on our dataset. DinoV2's self-supervised learning capabilities enable the model to uncover nuanced patterns, making it highly effective in distinguishing between real and fake images.

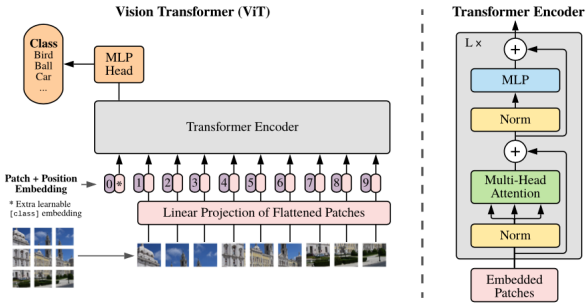


Fig. 2. Architecture of Vision Transformer

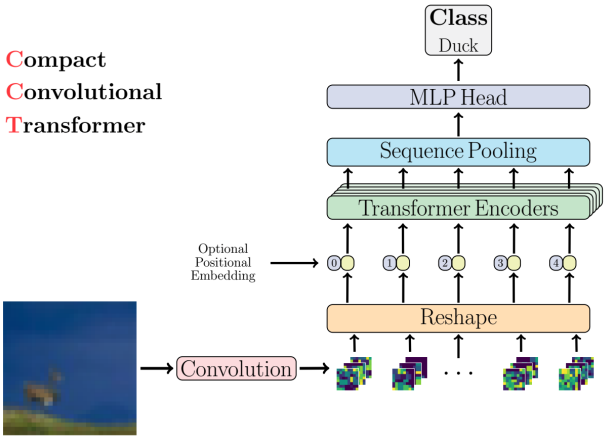


Fig. 3. Compact Convolutional Transformer

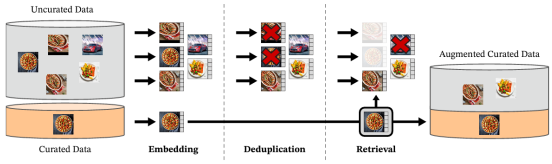


Fig. 4. DinoV2 Data Processing Pipeline

4) DCGAN: The Deep Convolutional GAN (DCGAN) was implemented with its discriminator network repurposed for real-fake image classification. The architecture of DCGAN demonstrates its effectiveness in generating realistic images by leveraging hierarchical feature learning. This characteristic makes it a robust framework for understanding the nuances of real and generated images.

The discriminator D in DCGAN is trained to differentiate between real images and those generated by the generator G. The objective of D is to maximize the probability of correctly classifying real and fake images, while G aims to minimize the probability of D correctly identifying its outputs as fake. This adversarial interaction is formalized as a minimax game,

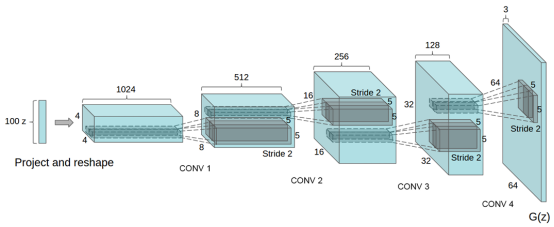


Fig. 5. Architecture of DCGAN generator

as described by Goodfellow et al. [20]. The GAN loss function is defined as:

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}[\log D(x)] + \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z)))] \quad (1)$$

Here, $D(x)$ represents the probability that x is a real image, and $D(G(z))$ is the probability that the output of the generator G is identified as real. This adversarial process enables G to generate images increasingly indistinguishable from real ones, enhancing the discriminator's ability to classify with greater precision.

IV. Implementation

For all models except the Generative Adversarial Network (GAN), the input image size was standardized to 224 pixels. Additionally, a learning rate scheduler (StepLR) was employed for all models to dynamically adjust the learning rate during training. The scheduler reduced the learning rate by a factor of 0.1 every five epochs.

A. Vision Transformer (ViT)

The Vision Transformer (ViT) was implemented with the cross-entropy loss function. For optimization, we employed the AdamW optimizer with an initial learning rate of 0.001. The training process spanned 10 epochs.

B. Compact Convolutional Transformer (CCT)

The Compact Convolutional Transformer (CCT) was also trained using the cross-entropy loss function as the criterion. Optimization was conducted using the Adam optimizer with an initial learning rate of 0.001. Training was conducted for a maximum of 30 epochs.
An early stopping mechanism was implemented, monitoring validation loss with a patience of 5 epochs to prevent overfitting. Training stopped at epoch 16.

C. Transfer Learning with DinoV2

The dataset for DinoV2 was preprocessed using its built-in data processor. Custom layers were added to the model to adapt it for the classification task. Transfer Learning techniques were employed to train the model. Initially, the model was trained end-to-end for 2 epochs. Subsequently, the base pre-trained layers were frozen, and only the custom output layers were trained for an additional 10 epochs using the Adam optimizer with a learning rate of 0.001.

D. Generative Adversarial Network (DCGAN)

The DCGAN model was trained using the Binary Cross-Entropy Loss (BCELoss) function. The Adam optimizer with a learning rate of 0.003 was employed during training. The model was trained for 100 epochs, after which the discriminator was utilized for validation and testing.

V. Results

In this section, we present the performance of four models—Vision Transformer (ViT), Compact Convolutional Transformer (CCT), DinoV2 with Custom Layers, and DCGAN—on the task of image classification. The models were evaluated based on their train accuracy, validation accuracy, test accuracy, and F1 score.

Model	Train Accuracy	Val Accuracy	Test Accuracy	F1 Score
ViT	1.00%	0.99%	0.96%	0.96087
CCT	1.00%	0.98%	0.95%	0.947285
DinoV2 + Custom Layers	0.94%	0.93%	0.92%	0.92
DCGAN		0.53%	0.49%	

TABLE I
Model Performance

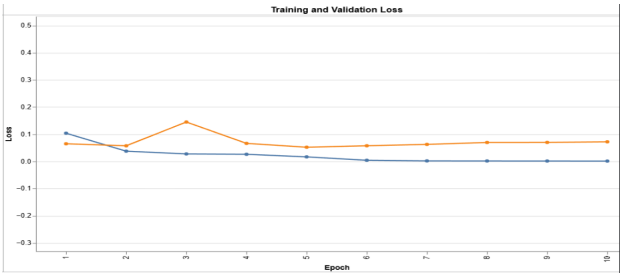


Fig. 6. Train and Validation Loss For ViT

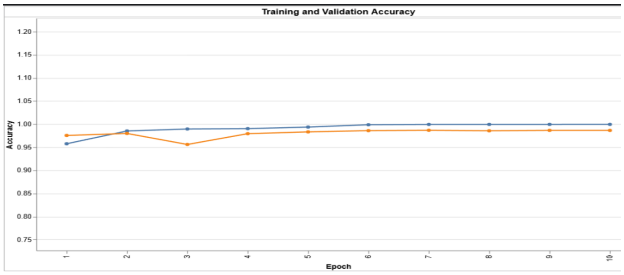


Fig. 7. Train and Validation Accuracy For ViT

VI. Future Work

While this study has focused exclusively on facial images, future research can expand to explore the detection of manipulated content in full-body images. Furthermore, the potential of self-supervised learning methods can be utilized more effectively to build versatile models that do not require separate training for different types of images. This approach would improve the adaptability and efficiency of fake image detection systems, particularly in handling diverse image datasets.

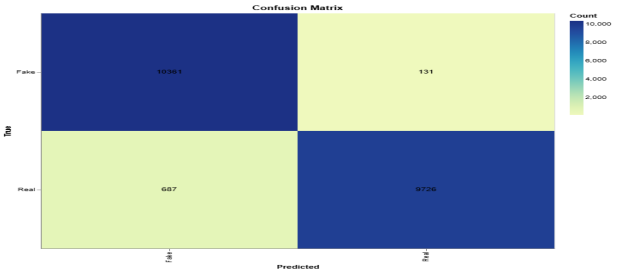


Fig. 8. Confussion Matrix For ViT

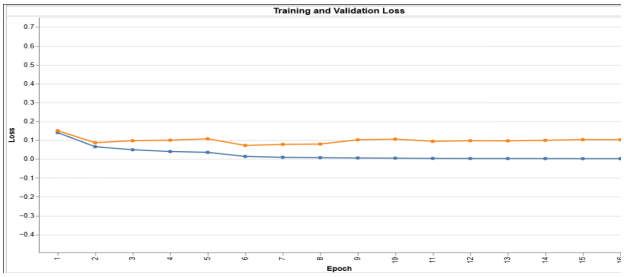


Fig. 9. Train and Validation Loss For CCT

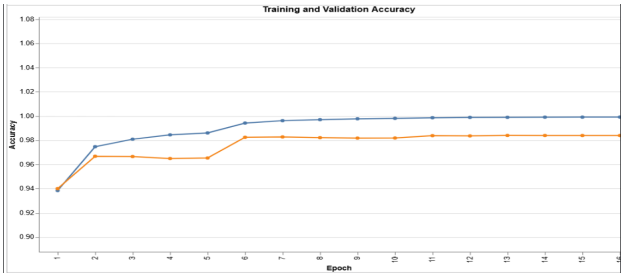


Fig. 10. Train and Validation Accuracy For CCT

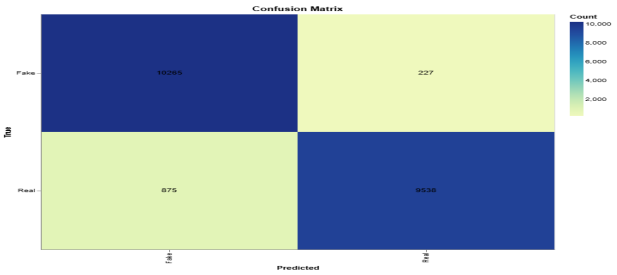


Fig. 11. Confussion Matrix For CCT

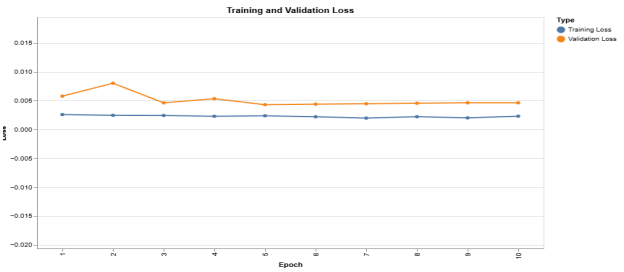


Fig. 12. Train and Validation Loss For DinoV2

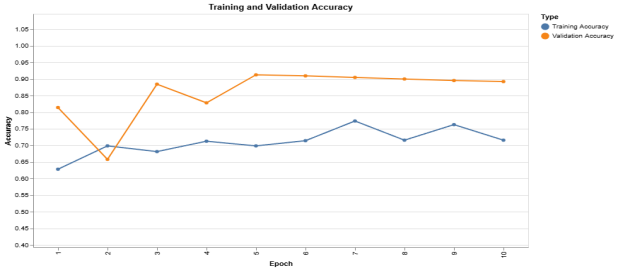


Fig. 13. Train and Validation Accuracy For DinoV2

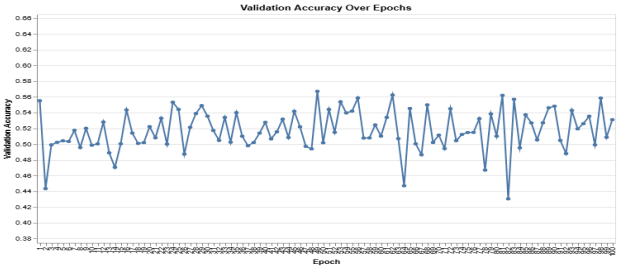


Fig. 14. DCGAN Descriminator Validation Accuracy

VII. Conclusion

This research addresses the pressing issue of detecting manipulated and AI-generated facial images by leveraging state-of-the-art techniques such as Vision Transformer (ViT), Compact Convolutional Transformer (CCT), and DINOv2 with custom layers. By focusing on a highly balanced dataset of facial images, our models demonstrate promising results in classifying real versus fake images. The incorporation of self-supervised learning methods lays the groundwork for more robust and scalable solutions. This work contributes to safeguarding individuals against digital harassment and ensures better security for facial biometric systems. Future research can further enhance these contributions by expanding the dataset and methodologies to encompass broader categories of images and refine the application of self-supervised learning approaches.

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